DEPARTMENT OF MECHANICAL ENGINEERING
FACULTY OF ENGINEERING AND ARCHITECTURE
MBM UNIVERSITY JODHPUR

Subject: 5ME42A(ME) – Manufacturing Technology

Tutorial # 2

Topic: Metal Working

1. The strength coefficient and strain-hardening exponent of a certain test metal are 40,000 lb/in^2 and 0.19, respectively. A cylindrical specimen of the metal with starting diameter = 2.5 in and length = 3.0 in is compressed to a length of 1.5 in. Determine the flow stress at this compressed length and the average flow stress that the metal has experienced during deformation.
2. Determine the value of the strain-hardening exponent for a metal that will cause the average flow stress to be 3/4 of the final flow stress after deformation.
3. A metal has a flow curve with strength coefficient = 850 MPa and strain-hardening exponent = 0.30. A tensile specimen of the metal with gage length = 100 mm is stretched to a length = 157 mm. Determine the flow stress at the new length and the average flow stress that the metal has been subjected to during the deformation.
4. A workpart with starting height h = 100 mm is compressed to a final height of 50 mm. During the deformation, the relative speed of the plattens compressing the part = 200 mm/s. Determine the strain rate at (a) h = 100 mm, (b) h = 75 mm, and (c) h = 51 mm.
5. A tensile test is carried out to determine the strength constant C and strain-rate sensitivity exponent m for a certain metal at 1000°F. At a strain rate = 10/sec, the stress is measured at 23,000 lb/in^2; and at a strain rate = 300/sec, the stress = 45,000 lb/in^2. (a) Determine C and m. (b) If the temperature were 900°F, what changes would you expect in the values of C and m?
6. A hot working operation is carried out at various speeds. The strength constant = 30,000 lb/in^2 and the strain-rate sensitivity exponent = 0.15. Determine the flow stress if the strain rate is (a) 0.01/sec (b) 1.0/sec, (c) 100/sec.

7. The strength coefficient = 550 MPa and strain-hardening exponent = 0.22 for a certain metal. During a forming operation, the final true strain that the metal experiences = 0.85. Determine the flow stress at this strain and the average flow stress that the metal experienced during the operation.
8. For a certain metal, the strength coefficient = 700 MPa and strain-hardening exponent = 0.27. Determine the average flow stress that the metal experiences if it is subjected to a stress that is equal to its strength coefficient *K*.
9. The strength coefficient = 35,000 lb/in^2 and strain-hardening exponent = 0.40 for a metal used in a forming operation in which the workpart is reduced in cross-sectional area by stretching. If the average flow stress on the part is 20,000 lb/in^2, determine the amount of reduction in cross-sectional area experienced by the part.
10. A tensile test is performed to determine the strength constant *C* and strain-rate sensitivity exponent *m* for a certain metal. The temperature at which the test is performed = 500°C. At a strain rate = 12/s, the stress is measured at 160 MPa; and at a strain rate = 250/s, the stress = 300 MPa. (a) Determine *C* and *m*. (b) If the temperature were 600°C, what changes would you expect in thevalues of *C* and *m*?